

Math 871 Topology I

Worksheet 4 solutions

1. Let $X = [0, 1] \times [0, 1] \subseteq \mathbb{R}^2$. Let $\mathcal{T}_{\text{sub}}$ be the subspace topology X inherits from $\mathbb{R}^2$, and let $\mathcal{T}_{<}$ be the dictionary order topology on X . Show that $\mathcal{T}_{\text{sub}}$ and $\mathcal{T}_{<}$ are incomparable topologies.

Note: to avoid confusion between points and intervals, let us write

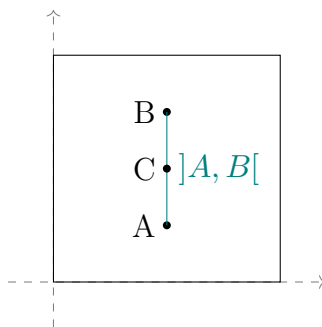
$$]A, B[:= \{x \in X \mid A < x < B\}$$

for the open interval from A to B .

Solution. First, we give an element of $\mathcal{T}_{<}$ that is not in $\mathcal{T}_{\text{sub}}$. Consider the points $A = (\frac{1}{2}, \frac{1}{4})$ and $B = (\frac{1}{2}, \frac{3}{4})$ in $\mathbb{R}^2$. The interval $]A, B[$ is a line segment, which can also be written as

$$]A, B[= \left\{ \frac{1}{2} \right\} \times \left] \frac{1}{4}, \frac{3}{4} \right[.$$

In particular, this is contained in the line $x = \frac{1}{2}$. Consider the point $C = (\frac{1}{2}, \frac{1}{2})$, which is in the open interval $]A, B[$.



In $\mathbb{R}^2$ with the standard topology, we can take the basis

$$\mathcal{B} = \{]a, b[\times]c, d[\mid a < b, c < d\}.$$

Any basic open $]a, b[\times]c, d[$ containing the point C will in particular have $a < \frac{1}{2} < b$ and $c < \frac{1}{2} < d$. Thus we can choose $e \in \mathbb{R}$ such that

$$\frac{1}{2} > e > \max\{a, 0\}$$

and the point $D = (e, \frac{1}{2})$ has

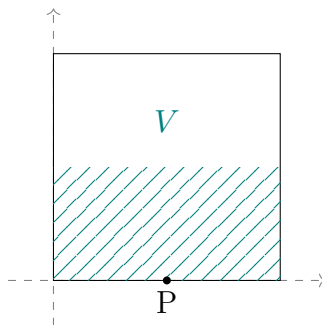
$$D \in]a, b[\times]c, d[\cap X \quad \text{but} \quad D \in D \notin]A, B[.$$

Now $\mathcal{B}_{\text{sub}} = \{]a, b[\times]c, d[\cap X \mid a < b, c < d\}$ forms a basis for $\mathcal{T}_{\text{sub}}$, and we just proved that no such basic open neighborhood of C is contained in $]A, B[$. We conclude that $]A, B[\notin \mathcal{T}_{\text{sub}}$, though by definition this is an open set in $\mathcal{T}_{<}$.

Now we give an element of $\mathcal{T}_{\text{sub}}$ that is not in $\mathcal{T}_{<}$. Now consider

$$V = [0, 1] \times \left[0, \frac{1}{2}\right] = \mathbb{R} \times \left[-1, \frac{1}{2}\right] \cap X \in \mathcal{T}_{\text{sub}}.$$

Fix the point $P = (\frac{1}{2}, 0)$, which is in V .



In the dictionary order topology, any open neighborhood of P must contain an interval of the form $]A, B[$ for some points $A < P < B$ in X . Write $A = (a_1, a_2)$ and $B = (b_1, b_2)$. Since it is not possible for $a_2 < 0$, then we must have $a_1 < \frac{1}{2}$; informally, A must be to the left of P . In particular, the interval $]A, P[\subseteq]A, B[$ contains the point $(a_1, 1)$, and is therefore not contained in V . We have thus shown that there are no open neighborhoods of P in $\mathcal{T}_{<}$ that are contained in V , so $V \notin \mathcal{T}_{<}$.

2. Let $Y \subseteq X$ and let $\mathcal{T}$ and $\mathcal{T}'$ be topologies on X such that $\mathcal{T}'$ is strictly finer than $\mathcal{T}$. Let $\mathcal{T}_Y$ and $\mathcal{T}'_Y$ be the corresponding subspace topologies on Y .

- (a) Prove that $\mathcal{T}'_Y$ is finer than $\mathcal{T}_Y$.

Solution. Let $U \in \mathcal{T}_Y$. By definition of the subspace topology, $U = V \cap Y$ for some open set $V \in \mathcal{T}$. Since $V \in \mathcal{T} \subseteq \mathcal{T}'$, we conclude that $U \in \mathcal{T}'_Y$.

- (b) Give an example showing that $\mathcal{T}'_Y$ might not be strictly finer than $\mathcal{T}_Y$.

Solution. Let $X = \mathbb{R}$, $Y = \mathbb{Z}$, and let $\mathcal{T}$ be the standard topology while $\mathcal{T}'$ is the discrete topology. Then $\mathcal{T}'$ is strictly finer than $\mathcal{T}$, but

$$\{n\} = (n-1, n+1) \cap \mathbb{Z} \in \mathcal{T}_Y$$

for all $n \in \mathbb{Z}$, and thus $\mathcal{T}_Y$ is the discrete topology. One can now check that $\mathcal{T}'_Y$ is also the discrete topology, or simply note that the discrete topology $\mathcal{T}_Y$ is already the finest topology of all, so $\mathcal{T}'_Y$ cannot be strictly finer.

3. Consider $\mathbb{R}$ under the included point topology with included point 0.

- (a) Find $\overline{\{0\}}$.

Solution. Given any point $x \in \mathbb{R}$ and any open neighborhood U of x , by definition of our topology $0 \in U$, and thus $U \cap \{0\} \neq \emptyset$. We conclude that $x \in \overline{\{0\}}$, and thus $\overline{\{0\}} = \mathbb{R}$.

(b) Find $\overline{\{1\}}$.

Solution. Note that $U = \mathbb{R} \setminus \{1\}$ is an open set in this topology, as it contains 0. We conclude that $\{1\}$ is closed.

4. Let Y be a subspace of X and S is any subset of Y . Show that

$$\overline{S}^Y = \overline{S}^X \cap Y.$$

Solution. ($\subseteq$) Since $\overline{S}^X$ is closed in X , then $\overline{S}^X \cap Y$ is closed in Y . Since $S \subseteq \overline{S}^X \cap Y$, then

$$\overline{S}^Y \subseteq \overline{S}^X \cap Y.$$

($\supseteq$) Let $y \in \overline{S}^X \cap Y$, and consider an arbitrary open neighborhood V of y in Y . By definition of the subspace topology, $V = U \cap Y$ for some $U \ni y$ open in X . Since $y \in \overline{S}^X$, then $U \cap S$ is nonempty. Since $S \subseteq Y$, we conclude that

$$V \cap S = (U \cap Y) \cap S = U \cap S$$

is nonempty. Since this holds for all open neighborhoods V of y in Y , we conclude that $y \in \overline{S}^Y$.

5. We say a topological space is T_0 if for every pair of distinct points x and y , there exists an open neighborhood U of x such that $y \notin U$ or there exists an open neighborhood V of y such that $x \notin V$. Note that if X is T_1 , then it is also T_0 . Given an example of a T_0 space that is not T_1 .

Solution. Let $X = \{a, b\}$ and

$$\mathcal{T} = \{\emptyset, X, \{a\}\}.$$

Note that this is a topology: for example, because it is the excluded point topology with excluded point b .

To show the space $(X, \mathcal{T})$ is T_0 , we just need to consider the only two distinct points, a and b . We can take the open neighborhood $U = \{a\}$ of a , and indeed $b \notin U$.

On the other hand, $\{a\}$ is not closed since its complement $\{b\}$ is not open. Since points are not closed, we conclude that this space is not T_1 .

6. A topological space X has the $T_{2\frac{1}{2}}$ property if for every pair of distinct points a and b , there are open sets U and V such that $a \in U$, $b \in V$, and $\overline{U} \cap \overline{V} = \emptyset$. Show that if S is a subspace of a $T_{2\frac{1}{2}}$ space X , then S is also $T_{2\frac{1}{2}}$.

Solution. Let $S \subseteq X$ and fix distinct points a and b in S . Since X is $T_{2\frac{1}{2}}$, we can find open sets U and V in X such that $a \in U$, $b \in V$, and $\overline{U} \cap \overline{V} = \emptyset$. Moreover, $A = U \cap S$ and $B = V \cap S$ are open sets in S with $a \in A$ and $b \in B$. Since $\overline{A}^S = \overline{A} \cap S$, then

$$\overline{A}^S \cap \overline{B}^S = (\overline{A} \cap S) \cap (\overline{B} \cap S) = (\overline{A} \cap \overline{B}) \cap S.$$

Now $A \subseteq U$ and $B \subseteq V$, so

$$\overline{A} \cap \overline{B} \subseteq \overline{U} \cap \overline{V} = \emptyset.$$

We conclude that $\overline{A}^S \cap \overline{B}^S = \emptyset$.